

The Bridge Experiment: Does an Informal $\leftrightarrow$ Formal Join Move End-Task Lean Performance?

Jack McCarthy

WikiLean · <https://wikilean.jackmccarthy.org>

with experiments executed by Claude Code agents under direction

Report, version 2 (**DRAFT**) — July 31, 2026 — post-review corrected edition

DRAFT — pre-adversarial-review edition. Two data slots are open: the blind-judge equivalence tables (§5) and the bare-union ablation (§6.3). Not for citation.

Preregistration: `docs/research/BRIDGE-EXPERIMENT.md`, commit `0d36f2664ab2ebb2c7b80b350d3c9bd820414335` (2026-07-16). Version 1 of this report (2026-07-25) received an external methodological review on 2026-07-27 (`docs/research/review/REVIEW-1.md`); we verified its factual claims independently (`docs/research/review/verification-of-review-1.json`: 29 confirmed, 1 partial, 1 refuted) and this edition incorporates the corrections. Version 1 remains in git history. All data, code, and run transcripts are preserved in the private repository `Deicyde/wikilean-bridge-experiment` (file map in §11). Every number below is recomputable from a file in that repository; the file is named where each table appears.

How to read this report. Every result carries one of three registers, marked at the section or table where it appears:

- **[PREREGISTERED-MODIFIED]** — a preregistered analysis executed with a named modification (the modification is stated where the tag appears).
- **[EXPLORATORY]** — not preregistered: all of Bridge v2 (retrieval, SorryDB, tool-use analysis) and the uncalibrated-judge grading.
- **[CORRECTIVE]** — a reanalysis finding of this edition (error repair, exposure audit, cluster-aware statistics).

Zero results in this report are cleanly confirmatory under the preregistered success criterion: that criterion was $D > E$ on *faithful@budget* (semantic equivalence to gold $\wedge$ typecheck) and $D \leq E$ on tokens-to-solve, and neither half was ever graded as specified (§3). The uncalibrated-judge section (§5) is the exploratory substitute for the missing equivalence leg.

1 Abstract

We test whether a curated dictionary between mathematical concepts and Mathlib declarations (the WikiLean “Brain”) improves language-model performance on formalization tasks, against controls holding the underlying corpora without the join. Results fall in three registers: one preregistered-but-modified comparison, a set of exploratory follow-ups, and the corrective reanalyses prompted by an external review; nothing here is cleanly confirmatory, because the preregistered semantic-equivalence endpoint was never graded as specified. Our primary metric is therefore the **grounded typecheck rate** — produced declaration, zero hallucinated citations, passing typecheck — which does not measure semantic correctness. On 100 post-Brain-index tasks (held out from the Brain’s indexes, but not — we later verified — from the formal-search arms’ sources), the bundled Wikibrain arm reaches 40.6% against 23.2% for the unjoined-tools control on the 69 completed pairs (McNemar 18/6, exact $p = 0.023$, risk difference +17.4pp [+4.1, +30.7]); after repairing the control’s 31 infrastructure-failed runs, the full-100 contrast is 42.0% vs. 30.0% ($p = 0.073$, RD +12.0pp [+0.2, +23.8]).

Join-specific attribution awaits a factorial ablation; the clean independent finding is the existence verifier, which cuts hallucinated citations to 6.8% against 17.7–21.2% in every other arm. Exploratory results: a post-hoc, benchmark-informed union-plus-manual agent obtained the highest rows we observed on MathlibQR-810 and MathlibMPR (declaration-clustered CIs exclude the single-arm baselines); on SorryDB, kernel-verified proving with and without the Brain is statistically indistinguishable (10 vs. 9 of 171, $p = 1.0$).

2 The system under test: the WikiLean Brain

The artifact everything below measures is the WikiLean “Brain”: a curated knowledge graph over mathematics that joins Wikipedia/Wikidata concepts, Mathlib4 declarations, and the external mathematical databases (nLab, LMFDB, the Stacks project, ProofWiki, and others) into one map. Its atom is a *cell*: when a Lean declaration, a Wikidata concept, an encyclopedia article, and a database page all denote the same mathematical object, they are *organs* of one cell rather than separate nodes — Mathlib’s `Module`, Wikidata’s “module” and its “vector space” are a single atom, because Mathlib has no `VectorSpace`; `Module` generalizes it. Mathlib’s folder hierarchy supplies the containment altitude, and everything known about a cell (Lean source, descriptions, licensed database snippets) is embedded on the cell itself. About 12,000 of Mathlib’s declarations are concept-joined this way; the Brain is a map of Mathlib, not the territory.

Agents reach it through an MCP server (`POST /mcp` on the live site) exposing eight tools. `brain_bridge` is the informal→formal entry point: free text in, existence-verified declarations out, each with its signature, import line, and bond quality. `brain_search` resolves labels and aliases to atoms; `brain_cell` returns the full atom card; `brain_transfer` makes the one-atom informal↔formal jump; `brain_neighborhood` walks the synapse graph; `brain_snippets` and `brain_filter` serve stored content and facet queries. The eighth, `decl_exists`, is different in kind: a batch existence check against the full declaration index — all of Mathlib, not just the joined subset — built to be called on every name an agent is about to cite. It matters below that `decl_exists` queries (effectively) the same doc-gen4 oracle our scorer uses to grade hallucinated citations: the bundle contains a tool aimed at a component of its own score, which is one reason this report no longer attributes effects to “the join” (§3, §9).

Why build such a thing? The working hypothesis behind WikiLean is that informal mathematics (what things are called and mean) and formal mathematics (what typechecks) are two corpora whose *join* is the scarce artifact: a language model already holds both corpora loosely, and the informal↔formal boundary is precisely where its hallucinations live. If that is right, then curating the join — names verified, concepts merged, bonds carrying provenance — should move performance on end tasks, not just on lookup. Whether it actually does is the question this report tries, and in this edition only partly manages, to answer.

3 Preregistered design and what actually ran

Hypothesis (operationalized). Since “necessary” is unprovable, the preregistration commits to three predictions. P1: an agent with the *joined* dictionary beats an agent given informal and formal search *separately*. P2: the bridge solves at lower cost. P3: the bridge lifts grounding (fewer hallucinated declaration citations), not just compile rate. The preregistered success criterion: $D > E$ on faithful@budget ($p < .05$) AND $D \leq E$ on tokens-to-solve.

The five arms. All five arms share the same model, prompt, and budgets within each phase; the only thing that differs between them is the tool manifest (`bench/run_bridge.py`, manifests in `bench/arms/`); see Table 1.

Two design facts the review surfaced, now stated up front. First, D is the only arm holding `decl_exists`, and hallucination-free citation is a conjunct of the primary metric, so D vs. E measures the *bundled Wikibrain package*, not the join per se (on the 100 fresh tasks D made 715 `decl_exists` call attempts, 682 successful). Second, arm E failed its manipulation check as an “actively consulting both corpora” control: across its 100 fresh runs it touched the informal tools 4 times, against arm B’s 345 — E behaved as a formal-search agent with a larger manifest (`verification-of-review-1.json`, arm-design section).

Arm	Tools	Intended to isolate
A no_tools	none (<code>-tools ""</code>)	floor / memorization
B informal	Wikipedia + nLab search/fetch, no Lean mapping	informal reasoning alone
C formal	loogle + ripgrep + source read over a Mathlib checkout	the LeanSearch-class status quo
D wikibrain	the Brain MCP (<code>brain_bridge/search/cell/transfer/neighborhood/snippets/filter + batch decl_exists</code>)	the join + the existence verifier (not separable in this design)
E B+C unjoined	B’s tools AND C’s tools, no join	the unjoined control

Table 1: The five preregistered arms. Identical model, prompt, and budgets within each phase; only the tool manifest differs.

Models per phase. Tier 1 uses `claude-haiku-4-5` (model id `claude-haiku-4-5-20251001`) in all five arms. Bridge v2 (retrieval and proving) uses `claude-sonnet-5` in all agent arms, per a pre-execution decision (`docs/research/BRIDGE-V2-BENCHMARKS.md`). One model per phase, one seed per (task, arm): a bridge that only helps some capability band would not be distinguished, and no run-to-run variance is measured anywhere.

Grading. The primary metric of this report is the **grounded typecheck rate**: the fraction of tasks where the run produced a declaration, cited zero names absent from a union oracle (`doc-gen4` declaration data $\cup$ verified renames; extractor documented in `bench/score_bridge.py`), and the declaration typechecked on the pinned toolchain.¹ It is a syntactic-validity + grounding measure. It does **not** establish that the produced statement formalizes the informal prompt: a declaration can pass all three legs while being weaker, stronger, vacuous, or about the wrong object. The preregistered `faithful@budget` (`BEq+` equivalence to gold) was never graded — the equivalence leg in `score_bridge.py` is a stub returning `None` — and §5 holds the exploratory substitute.

Task sets. Tier 1a is ProofNet#: 371 problems (341 eval + 30 prompt-freezing dev), pinned to `PAug/ProofNetSharp @a8da405fbd1e348a87445c2e562c747b7e26dc8f` (MIT; arXiv:2406.07222), graded on Lean v4.32.0-rc1 / Mathlib a33a5ccd. Tier 1b is the **post-Brain-index fresh set**²: 100 tasks built from theorems merged into Mathlib master between 2026-07-03 and 07-16, verified absent from the Brain’s declaration universe and node set — the indexes arm D serves (`bench/data/fresh_tasks.stats.json`) — and graded on Lean v4.33.0-rc1 / Mathlib 9944fe29. The set was **not** verified absent from the formal-search arms’ sources: arms C/E read a mutable local checkout (which stood at `61a5e4f338`, content through ~2026-07-10, during the runs) and queried the live unsnapshotted Loogle service. §4.4 measures the resulting exposure directly. A determinacy pre-screen was preregistered (exclude tasks two independent formalizers cannot pin down); what ran instead was a post-hoc double annotation (agreement 79%, $\kappa \approx 0.20$) defining a 74-task both-determinate subset — no tasks were excluded.

Budgets and execution. Every arm ran under a 10-minute wall clock (enforced) and a 30-turn budget that was **advisory only** — the prompt stated it, but the CLI passed no turn cap. Fresh-set overruns: C 50 runs, D 38, E 32 exceeded 30 turns, with maxima 80, 88, and 72. The fresh arms also ran in strictly sequential blocks on 2026-07-19 (A 01:50–02:08Z, B 02:09–03:56Z, C 03:58–04:27Z, D 04:29–04:54Z, E 04:56–05:17Z) — no interleaving, so arm is confounded with time of execution. Agents had no in-loop typechecker and were instructed to verify cited names through their tools.

¹Version 1 called this metric “success (folded)”. The rename follows the external review: a metric with no semantic-equivalence leg should not be called success.

²Version 1 called this set “contamination-proof” and “newer than every arm’s index”. Both phrasings are retracted (§9, concern 4).

The arm-E 429 incident and repair [CORRECTIVE]. Arm E’s fresh block died at its tail: rows fresh_069–099 (31 contiguous tasks) all errored with session-limit 429s — an infrastructure failure at the end of the sequential E block, not agent behavior. Version 1 scored these as “produced no declaration” and read them as evidence that unjoined tool volume is harmful; that reading is retired. On 2026-07-27 the 31 rows were rerun (`bench/analysis/rerun_E_fresh.py`) with E’s exact July-19 code path, on a read-only `git` archive extraction of 61a5e4f338 — the same tree content the original C/E runs saw. The repair surfaced a second condition-fidelity hazard worth recording: under concurrent cold starts, the CLI’s stdio MCP servers can silently fail to connect, yielding runs that *look* like completions but had zero tools. The repair driver detects this from the stream-json init event and forces such rows to error-and-retry rather than recording fake tool-less completions (commit 19a90209). Originals are archived in `bench/data/runs_E_fresh_429_archive/`; 14 of the 31 repaired rows became grounded-typecheck passes (`bench/analysis/bridge_summary_v2.json`, `v2_provenance`).

The complete execution table. Every preregistered component, its status, and the inferential consequence. (Sources: the preregistration; the claim-by-claim inventory in `verification-of-review-1.json`, prereg section.)

Preregistered component	Status	What actually happened	Inferential consequence
Five arms A–E, identical model/prompt/budget, tools-only difference	executed	471 rows/arm (<code>bench/data/runs/</code>)	design intact
Per-tool call logging	executed	<code>tool_calls_by_name</code> every row; full traces eval C/D/E + fresh (deviation 7)	enables §4.5 attribution
Tier 1a ProofNet# (371)	executed	371/arm, scored	contaminated by construction (arm A 59.8% on eval-341); paired deltas only
Tier 1b fresh set (~100)	executed	100 tasks	holdout guarantee narrower than claimed (Brain indexes only) — renamed post-Brain-index
Hallucinated-decl rate	executed	all tables	the cleanest surviving mechanism signal
McNemar D-vs-E / D-vs-C + discordant tables	executed	§4	run on grounded typecheck, not the preregistered endpoint
10-min wall clock	executed	600s timeouts	—
Pre-campaign API requirements 1–8	executed	implemented before campaign	—
30-turn budget	modified	advisory only; overruns C 50/D 38/E 32, max 88	budget is uncontrolled; §4.3 sensitivity analysis
Determinacy pre-screen (exclude)	modified	post-hoc dual annotation (79%, $\kappa \approx 0.20$); 74-task subset, none excluded	subset analysis, not screened population
Arm-D production server	modified	production Worker code served locally over shipped shards (deviation 2)	none material
Single-shot elaboration grading	modified	persistent REPL, same pins (deviation 3)	none material
≤ 4 typecheck calls in-loop	modified	no in-loop typechecking at all (deviation 4)	halluc rate is a pure grounding measure
Skill/ToolSearch leak	modified	sealed before eval; 120 dev runs quarantined (deviation 6)	dev split discarded, eval clean
Per-arm cost distributions faithful@budget (BEq+ equivalence)	modified	means only (USD + wall-clock)	descriptive
LLM judge + 50-item human calibration	not executed (campaign)	grader is a stub; zero judge files in campaign	primary endpoint missing; no confirmatory claim possible
tokens-to-solve (half the success criterion)	not executed	harness existed unused; an <i>uncalibrated</i> blind judge pass was launched post-review (§5)	§5 is exploratory only
3 reseeds + pass@k curves	not executed	never computed (tokens recorded per row)	P2 untested as specified
Second model class on primary set	not executed	one run per (task, arm) everywhere	no variance estimate; single-seed caveat on every number
		Tier-1 all-Haiku; v2 all-Sonnet on different benchmarks	capability-band generality unknown

Preregistered component	Status	What actually happened	Inferential consequence
Preregistered success criterion	not executed	neither half graded as specified	zero confirmatory results
Tier 2 as specified (FATE-H + MPR-Prop proving, reflection loop, arms A/C/D/E) PutnamBench	not executed	replaced by exploratory v2: QR/MPR retrieval + one-shot SorryDB, arms N/F/W/WF	§§6–7 are exploratory, not the preregistered Tier 2
Tier 3a offline Erdős set	not executed	—	—
Tier 3b live Erdős queue	not executed	—	—

4 Tier 1: statement autoformalization (claude-haiku-4-5)

[PREREGISTERED-MODIFIED] — the preregistered paired design, graded on grounded typecheck rate instead of faithful@budget, one model, one seed, advisory turn budget. Corrected tables below are from `bench/analysis/tier1_reanalysis.{json,md}` and `bench/analysis/part1_fresh100_v2.{json→bridge_summary_v2.json,md}`; all rates carry Wilson 95% CIs.

4.1 Tier 1a — ProofNet# eval split (n=341/arm)

arm	grounded typecheck	Wilson 95% CI
A none	204/341 = 59.8%	[54.5, 64.9]
B wiki	198/341 = 58.1%	[52.8, 63.2]
C formal	218/341 = 63.9%	[58.7, 68.9]
D wikibrain	219/341 = 64.2%	[59.0, 69.1]
E B+C unjoined	208/341 = 61.0%	[55.7, 66.0]

The arms sit within a few points of each other; the v1 D-vs-E McNemar (63 vs. 51 discordant over the 371 folded pairs, exact $p = 0.30$) is an underpowered null. Arm A at ~60% with zero tools is substantial ProofNet memorization and was the motivation for Tier 1b. The typecheck-only row still *anti-correlates* with grounding (E led typecheck at 83.3% while trailing on hallucinations in the v1 tables) — reproducing TheoremGraph’s typecheck-is-not-a-signal finding at $15\times$ their n . Tier-1a is contaminated by construction; we report it for the paired deltas and as the memorization control, never alone.

4.2 Tier 1b — the post-Brain-index fresh set (n=100/arm)

Three analyses, in order of increasing completeness. First, the campaign as it actually ran, with E’s 31 infrastructure-dead rows counted as failures (intention-to-treat over the snapshot; `tier1_reanalysis.md` §2):

arm	grounded typecheck	Wilson 95% CI	infra errors
A	20/100 = 20.0%	[13.3, 28.9]	0
B	22/100 = 22.0%	[15.0, 31.1]	0
C	25/100 = 25.0%	[17.5, 34.3]	0
D	42/100 = 42.0%	[32.8, 51.8]	0
E	16/100 = 16.0%	[10.1, 24.4]	31

E’s row conflates agent behavior with a rate-limit outage; the D-vs-E McNemar on this table is the error-inflated figure v1 headlined, and it now appears only in §9. The two contrasts *unaffected* by the outage (their arms had zero errors): **D vs. C 28/11 discordant**, exact $p = 0.0095$; **D vs. A 29/7**, $p = 0.0003$ (100 pairs each).

Second, the completed-pairs analysis — the 69 tasks arm E actually ran — which is this report’s headline Tier-1 contrast (`tier1_reanalysis.md` §3, §6):

arm	grounded typecheck (n=69)	Wilson 95% CI
A	10/69 = 14.5%	[8.1, 24.7]
B	12/69 = 17.4%	[10.2, 28.0]
C	12/69 = 17.4%	[10.2, 28.0]
D	28/69 = 40.6%	[29.8, 52.4]
E	16/69 = 23.2%	[14.8, 34.4]

D vs. E: McNemar 18/6 discordant, exact $p = 0.023$; risk difference +17.4pp, 95% CI [+4.1, +30.7] (paired Wald). Also D vs. C 21/5, $p = 0.0025$; D vs. A 22/4, $p = 0.0005$ on these pairs.

Third, the post-repair full 100 — all 500 fresh rows completed, the 31 repaired E rows scored with the identical pipeline (bench/analysis/part1_fresh100_v2.md, from bridge_summary_v2.json):

arm	grounded typecheck (n=100)	Wilson 95% CI
A	20/100 = 20.0%	[13.3, 28.9]
B	22/100 = 22.0%	[15.0, 31.1]
C	25/100 = 25.0%	[17.5, 34.3]
D	42/100 = 42.0%	[32.8, 51.8]
E	30/100 = 30.0%	[21.9, 39.6]

D vs. E post-repair: McNemar 25/13, exact $p = 0.073$; RD +12.0pp, 95% CI [+0.2, +23.8]. D vs. C ($p = 0.0095$) and D vs. A ($p = 0.0003$) are unchanged — only E’s cells moved. Stated plainly: with the infrastructure failures repaired, the bundled-Wikibrain-vs-unjoined contrast on the full set is a directionally consistent but marginal effect (the RD interval barely excludes zero; the McNemar p does not reach 0.05); the completed-pairs contrast is significant. And E, repaired, is an unremarkable mid-pack arm (30%, statistically indistinguishable from C, $p = 0.42$): version 1’s “unjoined tool volume can be actively harmful” was an infrastructure artifact and is retired.

Arm A’s collapse from 59.8% (1a) to 20% here remains the cleanest memorization quantification in the study.

4.3 Turn-budget sensitivity [CORRECTIVE]

The 30-turn budget was advisory (§3), so we recompute restricted to pairs where **both** arms stayed within 30 turns (snapshot outcomes; tier1_reanalysis.md §5). D vs. E: $n = 45$ pairs, D 46.7% [32.9, 60.9] vs. E 24.4% [14.2, 38.7], 16 discordant, $p = 0.021$ — but this error-inclusive version counts E’s 429 rows (turns=1) as within-budget failures. The honest completed-only version is underpowered: $n = 27$ pairs, D 44.4% [27.6, 62.7] vs. E 40.7% [24.5, 59.3], 7 discordant, $p = 1.0$. D vs. C within budget: $n = 35$, D 54.3% [38.2, 69.5] vs. C 25.7% [14.2, 42.1], 12 discordant, $p = 0.0063$. The budget-restricted evidence is consistent in direction and, for D-vs-E, inconclusive once both restrictions are applied at once.

4.4 Exposure strata: what the formal-search leak actually touched [CORRECTIVE]

The review found the fresh golds were not held out from arms C/E’s sources. We measured it (bench/analysis/fresh_exposure.{json,md}; read-only git archive of the pinned tree 61a5e4f338, content date 2026-07-10). Of the 100 tasks: the gold’s full dotted name appears verbatim somewhere in the tree for **37**; the basename appears as a declaration header anywhere for **64**; the basename appears as a declaration header **in the task’s own module file** for **51** — the exposure basis used below; and the gold’s commit is an ancestor of the pin for 49.³

Split outcomes (snapshot basis: E’s 429 rows count as failures, so E’s unexposed cell is heavily infrastructure-depressed; the attempted-only rows exclude them):

Exposed stratum ($n = 51$):

³Exposure is measured on tree bytes, not merge metadata: two tasks (fresh_037, fresh_054) are exposed despite post-pin added-commits — fresh_054’s full dotted name already sits in its module file at the pin — which is why 51 exposed $\neq$ 49 merged-before-pin.

arm	grounded typecheck	Wilson 95% CI
A	9/51 = 17.6%	[9.6, 30.2]
B	8/51 = 15.7%	[8.2, 28.0]
C	11/51 = 21.6%	[12.5, 34.6]
D	24/51 = 47.1%	[34.1, 60.5]
E	12/51 = 23.5%	[14.0, 36.8]
E attempted-only	12/44 = 27.3%	[16.4, 41.9]

Unexposed stratum ($n = 49$):

arm	grounded typecheck	Wilson 95% CI
A	11/49 = 22.4%	[13.0, 35.9]
B	14/49 = 28.6%	[17.8, 42.4]
C	14/49 = 28.6%	[17.8, 42.4]
D	18/49 = 36.7%	[24.7, 50.7]
E	4/49 = 8.2%	[3.2, 19.2]
E attempted-only	4/25 = 16.0%	[6.4, 34.6]

McNemar by stratum: D vs. E exposed 15/3, $p = 0.0075$; **D vs. E unexposed 17/3**, $p = 0.0026$. D vs. C exposed 18/5, $p = 0.0106$; D vs. C unexposed 10/6, $p = 0.45$.

The reading has two honest halves. Against E, the leak cannot explain D’s edge: the leak’s direction favors C/E (direct source access to the golds; D’s Brain indexes genuinely predate every gold), and D’s advantage over E is *strongest exactly where there was nothing to leak* — 36.7% vs. 8.2% on unexposed tasks (with the caveat that E’s unexposed denominator is the most outage-damaged stratum; even attempted-only, 36.7% vs. 16.0%). Against C the picture is weaker: D’s edge concentrates in the exposed stratum and is a null ($p = 0.45$) on unexposed tasks — consistent with D’s advantage over pure formal search depending partly on targets a formal-search agent could in principle have found in-tree but didn’t.

4.5 Hallucinated citations and trace attribution

Hallucination rates on the fresh set, all 100 rows per arm (post-repair; recomputed for this edition from `bench/data/runs/{A..E}/fresh_*.json` with `score_bridge.py`’s union oracle):

arm	halluc-decl rate (cited names)	runs w/ ≥ 1 hallucination	Wilson 95% CI (runs, $n = 100$)
A	94/443 = 21.2%	54	[44.3, 63.4]
B	76/429 = 17.7%	48	[38.5, 57.7]
C	108/517 = 20.9%	49	[39.4, 58.7]
D	32/472 = 6.8%	23	[15.8, 32.2]
E	107/505 = 21.2%	49	[39.4, 58.7]

The citation-level rates are descriptive (citations cluster within runs); the run-level proportions carry the CIs. This is the study’s most robust effect: it survives the repair unchanged (D 6.8% vs. 17.7–21.2%), it is larger off-distribution than on (Tier-1a: D 5.9% vs. 10.1–11.3%), and the traces explain its mechanism.

Trace attribution [EXPLORATORY — eval split only] (deviation-7 telemetry covers eval arms C/D/E; `bench/trace_analysis.py`): 98–100% of traced tool-arm runs cite at least one declaration that visibly surfaced in a tool result, and only 10–17% of citations never touched the tools. 35% of arm-D eval runs (116/329) cite a declaration that came out of a `brain_bridge/brain_transfer` result — English in, formal name out (an undercount; result heads truncate at 200 chars). The sharpest signal is among *hallucinated* citations: the checked-and-cited-anyway rate is 35% for C and 30% for E but **13% for D** — binary verification disciplines the model where the fuzzy neighborhoods of loogle and grep let it fool itself.

4.6 The existence verifier as an independent finding

We promote this from mechanism-footnote to first-class result, following the review’s own suggestion. What the data support: giving an agent a **batch, binary, oracle-backed existence check** (plus the instruction

to use it on every name) collapses hallucinated citations by roughly $3\times$ across two task distributions, and the effect concentrates precisely on the citations the agent had already “checked” by search. What the data do not yet support: how much of D’s grounded-typecheck edge is this verifier versus the concept join — D is the only arm holding it, and no-halluc is a conjunct of the metric. The 2×2 factorial (join $\times$ decl_exists; roadmap §10.1) is the experiment that separates them. Note the circularity honestly: `decl_exists` and the scoring oracle draw on the same doc-gen4 index, so “fewer hallucinations” partly means “the arm could query the grader’s oracle” — which is itself the product claim, but must be named.

Cost (descriptive only). Campaign means per task, folded over 471 runs/arm as run (pre-repair; E’s 31 error rows cost ≈ 0): A \$0.034, B \$0.048, C \$0.140, D \$0.121, E \$0.128; mean wall-clock C 116s, D 89s, E 106s. The preregistered tokens-to-solve test was never computed, so P2 remains untested; v1’s “cheaper and better” framing is withdrawn to this descriptive note.

5 Semantic faithfulness (uncalibrated judge) [EXPLORATORY]

The preregistered primary endpoint — equivalence to the gold statement — was never graded during the campaign (§3). Following the review, a blind LLM-judge pass over all 500 fresh outputs (5 arms $\times$ 100 tasks, including the 31 repaired E rows) was launched on 2026-07-27 (`bench/analysis/judge_fresh_run.py`). Protocol: judge = `claude-sonnet-5` (deliberately a different, stronger model than the Haiku subjects), no tools, one turn, prompt containing only {informal statement, gold, produced declaration}; every rendered prompt is scanned for arm-revealing substrings and the driver aborts on any hit; no-output rows are judged not-equivalent by definition; a fixed 50-item seed-stratified subset (seed 20260727, 10 per arm) is re-graded for self-consistency. **The judge is uncalibrated:** the preregistered 50-item human calibration remains undone, so every number in this section is exploratory and none can rescue the missing confirmatory endpoint.

Pending: blind-judge equivalence tables (grading in progress)

[[SLOT:JUDGE]] — to be filled from `bench/analysis/judge_fresh_summary.json` when the running pass completes (as of this draft, arms A–D are graded and E is in progress). The tables to land here, per `judge_fresh_summary.py`:

1. Per-arm strict and evaluated equivalence rates on fresh-100, Wilson 95% CIs.
2. The conjunction (grounded typecheck $\wedge$ judge-evaluated-equivalent) per arm — the closest available analogue of faithful@budget.
3. Exact-binomial McNemar D-vs-E, D-vs-C, D-vs-A on judge-evaluated and on the conjunction, full-100 and completed-69.
4. Self-consistency: strict/evaluated agreement on the 50-item re-grade, plus judge cost/error census.

One grading-fidelity deviation is pre-declared: for fresh tasks the judge sees `gold_context` + `gold_formal` (the `variable/open` binders the statement needs); this is arm-independent.

6 Third-party retrieval: MathlibQR-810 and MathlibMPR (claude-sonnet-5) [EXPLORATORY]

Bridge v2 replaces judge-dependent grading with graders that predate the arms: third-party expert gold labels here, the Lean kernel in §7. Data: `frenzymath/LeanSearch-v2 @94f4888cbaf9` (CC BY 4.0; arXiv:2605.13137), copied under `bench/v2/data/`. Four arms: N (no tools), F (loogle + `decl_grep` + `decl_read`), W (the Brain MCP), and WF (W $\cup$ F **plus** `bench/v2/AGENT_MANUAL.md` prepended to the prompt). Scoring is `bench/v2/score_retrieval.py`, exact full-name match; run rows with full gzipped stream-json transcripts in `bench/v2/runs/agent/`.

WF is a post-hoc, benchmark-informed condition — every WF number below carries that label. The timeline (commit times, verified): N/F/W results committed 22:27 on 07-24; the manual written 00:30 on 07-25; WF results 01:15. The manual distills measurements taken **on the same evaluation queries**

WF was then scored on — its bracketed figures (per-style nDCG values, the 63% brain_cell failure rate, the 143/810 format failures, per-tool call counts) are the N/F/W/system-mode eval-set aggregates, reproduced exactly (`verification-of-review-1.json`, manual-timing section). WF is development evidence about what a maximally equipped, briefed agent can do — not an untouched test result, and not a defensible SOTA comparison. Version 1 additionally misattributed the 63% brain_cell figure to “the Tier-1 traces”; it is the v2 W-arm eval figure (482 calls, 63.3% failed). Corrected here and in §9.

6.1 MathlibQR fair-810 — concept retrieval, declaration-clustered [CORRECTIVE statistics]

The 810 query rows are paraphrase styles of only **171 distinct gold declarations** (2–6 rows each), so v1’s row-level McNemars overstated the effective n . All inference is now at the declaration level (`bench/analysis/retrieval_clustered.md`; cluster bootstrap $B = 10,000$, seed 20260727):

system	R@10 (row)	R@10 95% CI (decl-cluster boot)	nDCG@10	nDCG 95% CI
published: TheoremGraph	0.775	—	0.548	—
published: LSv2 retriever+reranker	0.780	—	0.623	—
system-mode wikibrain (one brain_bridge call, no LLM)	0.036	—	0.031	—
agent N	0.633	[0.581, 0.684]	0.598	[0.547, 0.647]
agent F	0.831	[0.792, 0.868]	0.790	[0.750, 0.829]
agent W	0.816	[0.767, 0.862]	0.781	[0.731, 0.827]
agent WF (post-hoc, benchmark-informed)	0.885	[0.849, 0.919]	0.839	[0.801, 0.874]

Declaration-level paired contrasts: **F – W is a null** (R@10 +0.015, 95% CI [−0.028, +0.059], Wilcoxon $p = 0.26$; nDCG +0.010 [−0.029, +0.048]) — the two toolkits are statistically tied, with different textures (W wins the special_case style, nDCG 0.523 vs. 0.384; F wins the Lean-syntax styles; grep cannot find what source text never names). WF – F: +0.054 [+0.027, +0.082], Wilcoxon $p = 0.0017$; WF – W: +0.069 [+0.032, +0.108], $p = 0.0003$ — the briefed union arm beats both single arms even under clustering, with the post-hoc label attached.⁴

The system-mode row is the headline API deficiency and survives every correction: one brain_bridge call scores 0.036 because the free-text entry is a label/alias resolver, not a semantic retriever — while an agent *iterating* over the same API reaches 0.816. The content is in the graph; the single-shot entry point is the gap. Agent N’s 0.633 includes 143/810 rows scored 0 for format non-compliance.

6.2 MathlibMPR — premise retrieval (69 post-cutoff PR theorems)

One task per PR (the review’s clustering concern does not apply here — see §9); task-level bootstrap CIs:

system	group-recall@10	95% CI (task boot)
published: LeanSearch-v2 reasoning	0.461	—
published: DIVER	0.380	—
published: TheoremGraph concept-tuned	0.165	—
system-mode wikibrain	0.000	—
agent N	0.203	[0.131, 0.282]
agent W	0.272	[0.196, 0.354]
agent F	0.453	[0.365, 0.540]
agent WF (post-hoc, benchmark-informed)	0.557	[0.472, 0.642]

Two readings. A generic Sonnet agent with grep and loogle **matches the specialist premise retriever** (0.453 vs. 0.461) — a notable baseline on its own, untouched by any of this study’s confounds. And W trails F

⁴Version 1’s row-level McNemars ($n = 810$ treated as independent: F–W 78/66 discordant $p = 0.36$; WF–F 71/27 $p = 1.0 \times 10^{-5}$; WF–W 83/27 $p = 8.3 \times 10^{-8}$) are **uncorrected** for the declaration clustering — the 144 F/W discordant rows come from only 76 declarations — and are superseded by the clustered tests above.

by 18pp: the preregistered concept $\neq$ premise boundary (TheoremGraph’s negative transfer), now measured on our own tools, giving **brain_premises** (BRIDGE-ISSUES #7) a quantified target. The WF – F contrast is **marginal**: $+0.104$, 95% CI $[+0.007, +0.201]$, sign test $p = 0.029$, Wilcoxon $p = 0.049$ — the interval nearly touches zero, and the post-hoc label applies.

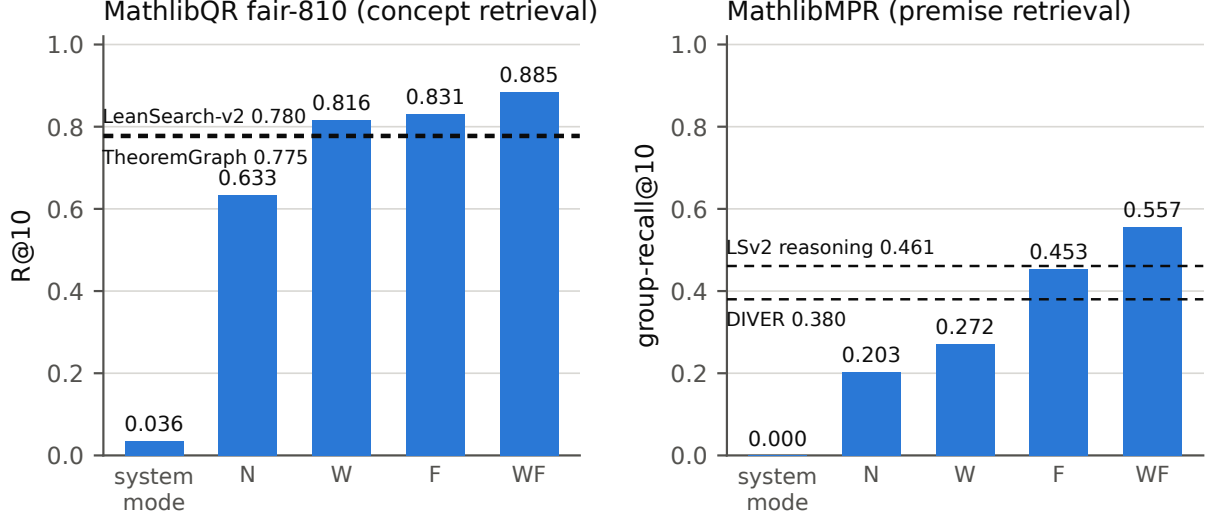


Figure 1: Retrieval point estimates by system, with published anchors as dashed reference lines. Left: MathlibQR fair-810 R@10 (anchors: TheoremGraph 0.775, LeanSearch-v2 retriever+reranker 0.780). Right: MathlibMPR group-recall@10 (anchors: LeanSearch-v2 reasoning 0.461, DIVER 0.380). System-mode is a single **brain_bridge** call with no LLM; N/W/F/WF are Sonnet agent arms; WF is the **post-hoc, benchmark-informed** condition (§6). The point estimates are unchanged from version 1 (data: `bench/v2/score_retrieval.py`); the confidence intervals and significance tests were *redone at the declaration level for this edition* (§6.1 declaration-cluster bootstrap, §6.2 task bootstrap) and are reported in the tables above, not drawn in this figure.

6.3 Tool use under the manual

The census (pooled QR+MPR): F averages 2.2 calls/run; W 3.5 (decl_exists 1,251 + brain_bridge 608 — verify-then-cite made mechanical); WF 4.6 in a genuine dual-toolkit mix (decl_grep $\sim 1.5k$, decl_exists $\sim 0.9k$, brain_bridge $\sim 0.7k$, decl_read $\sim 0.5k$, loogle $\sim 0.4k$). Under the manual, brain_cell misuse (63% failure rate in the v2 W-arm runs) falls to a single call in WF. These are development observations about the same eval set the manual was tuned on, and are labeled accordingly.

Pending: bare-union (U) ablation (queued)

*[[SLOT:UNION]] — the bare-union U arm ($W \cup F$ tools, no manual; **MANUAL_ARMS** blanked in `bench/v2/run_agent.py`) separates the union effect from the briefing effect. Planned table on completion: N / F / W / U / WF on both benchmarks, declaration-clustered (QR) and task-bootstrap (MPR) CIs, with U-vs-F and WF-vs-U paired contrasts. Until it lands, “union tools” and “benchmark-informed manual” remain unseparated in WF.*

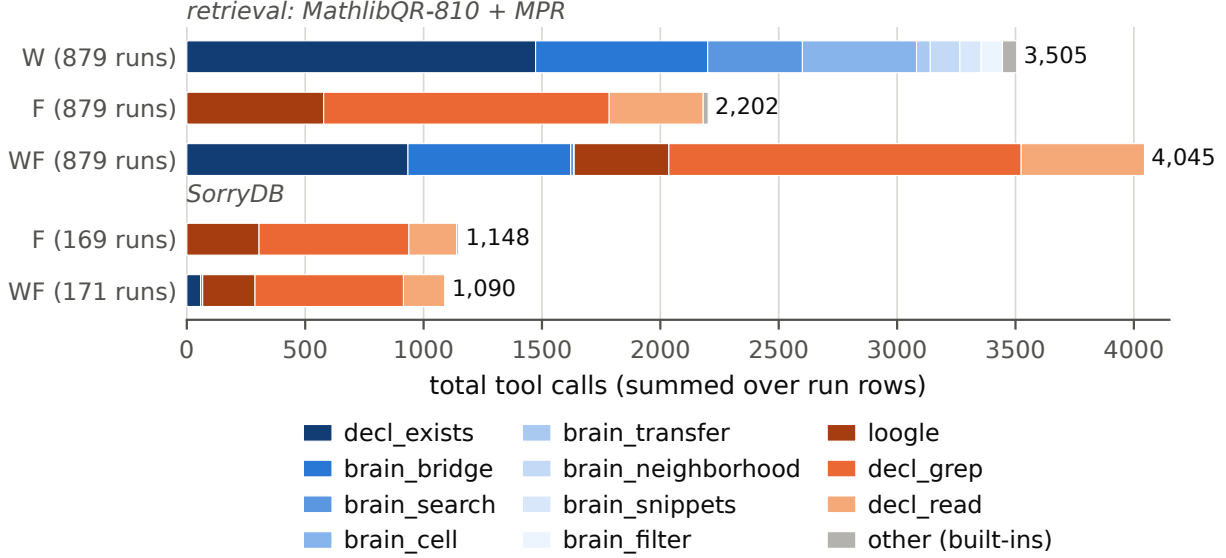


Figure 2: Tool composition by arm: total calls summed over run rows (retrieval pools QR-810 and MPR, 879 runs/arm; SorryDB ~170 runs/arm). Blues are Brain tools, oranges formal-search tools, gray the CLI’s built-ins. `decl_exists` dominates every Brain-bearing arm; under the manual (WF — the **post-hoc, benchmark-informed** condition, §6) the Brain toolkit contracts to `brain_bridge` + `decl_exists` (`brain_cell`: 482 calls in W, one in WF); and on SorryDB the WF agent routes 94% of its calls to formal search (§7). This telemetry is unchanged by this edition’s corrections. Data: `transcript_stats.tool_calls_by_name` summed over `bench/v2/runs/`.

7 SorryDB: kernel-graded proving in live repositories (claude-sonnet-5) [EXPLORATORY]

SorryDB (arXiv:2603.02668, <https://sorrydb.org>) serves unsolved `sorry`s from real Lean repositories; success is the repo *building* with your proof — no judge anywhere, nothing to memorize. Snapshot: the SorryDB-2601 1,000-task evaluation split.

Subset construction and the snapshot-rot finding. Disk permitted one repository build at a time, so we sampled repositories rather than tasks. The original freeze — 8 repositories, 164 tasks — lost **74 of 164 tasks (45%)** in the ~six months since the snapshot: pinned commits of `GlimpseOfLean`, `LeanCourse25`, and most Foundation and SemicircleLaw tasks no longer exist on GitHub (history rewritten; unreachable commits deleted). No retrieval route worked, and detection is subtle: GitHub’s archive endpoint answers preflight HEAD for dead commits with HTTP 200, so a liveness check that does not actually fetch reports the pins healthy. Methodological finding: benchmarks pinned to live repositories rot at the commit level, and the rot is invisible unless every pin is re-fetched before use. The refreeze (`bench/v2/sorrydb_prep.py`, commit `db679e8d`) keeps only verified-fetchable pins: **171 tasks across 10 repositories**.

Protocol. Build-free, one-shot: each row carries the goal state plus a ± 60 -line window at the pinned commit; the agent (600s, arms N/F/WF as in §6) outputs one replacement for the `sorry`, with an explicit honest-abstention rule. Verification (`bench/v2/verify_sorrydb.py`) clones at the pin, splices, rebuilds the module; success = exit 0 with no `sorry`/axiom warning.

Results — verdict-complete. All 203 frozen-task candidate proofs now have kernel verdicts (commit `9682b3b1`; the 8 `curve25519` stragglers of v1 verified as 0 proved). Recomputed from `bench/v2/runs/sorrydb/verify.jsonl` × the run rows, $n = 171$ tasks/arm:

	N no tools	F formal	WF union+manual (post-hoc)
run rows present	168	169	171
no output (timeout / no lean block)	70	15	11
honest give-up (literal <code>sorry</code>)	40	83	86
candidate proofs submitted	58	71	74
proved (kernel)	2	9	10
failed	50	55	57
unspliceable	5	7	7
verify timeout	1	0	0
proved / 171	1.2%	5.3%	5.8%
Wilson 95% CI	[0.3, 4.2]	[2.8, 9.7]	[3.2, 10.4]
repo-clustered boot 95% CI	[0.0, 2.7]	[0.0, 10.7]	[0.0, 12.2]
proved / candidates	3.4%	12.7%	13.5%
total cost (USD)	\$58.97	\$102.76	\$108.87
cost per proved	\$29.48	\$11.42	\$10.89
mean wall-clock / task	120 s	198 s	183 s

(3 N and 2 F tasks have no run row — runner losses. Distinct sorries proved across arms: 13; N’s are a strict subset of both F’s and WF’s; $WF \cap F$ 6, WF-only 4, F-only 3. Proofs concentrate in 3 of the 10 repos: PersistentDecomp, LeanAPAP, MiscYD.)

WF vs. F is statistically indistinguishable. The point difference is one proof (10 vs. 9 of 171): task-level exact McNemar 4/3, $p = 1.0$; repo-clustered bootstrap difference +0.58pp, 95% CI [+0.0, +1.85pp], with **34% of resamples showing no difference or worse (retrieval_clustered.md §4)**. SorryDB provides essentially no evidence about the Brain: WF routed 94% of its tool calls to formal search here, and no W-only or bare-union arm was run. What the experiment does show is the tools-vs-no-tools gap — N burns its budget (70 of 168 rows produce nothing, mostly 600s timeouts) while tool arms triple-to-quintuple the prove rate and double the honest-abstention rate — though with only 10 repo clusters even that deserves caution. The cost-per-proved row is retained as bookkeeping; with 2/9/10 successes it is unstable, and v1’s “adding the join never made the task more expensive per success” is deleted (§9). The published SorryDB agentic best ($\approx 30\%$) is not comparable: different task subset (research-frontier-weighted) and materially different protocol (published agents iterate against the build; ours drafted one-shot).

8 What survives and what does not

Each retired v1 claim, paired with its v2 replacement.

v1 claim (retired)	v2 replacement (supported)
“The join carries the effect” (fresh McNemar $p < 0.0001$)	The bundled Wikibrain condition outperforms the tested controls: completed-69 D 40.6% vs. E 23.2% ($p = 0.023$, RD +17.4pp [+4.1, +30.7]); post-repair full-100 42.0% vs. 30.0% ($p = 0.073$, RD +12.0pp [+0.2, +23.8]); D vs. C $p = 0.0095$. Join-specific attribution awaits the 2×2 factorial.
“42% success”	“42% grounded typecheck rate ” — no semantic-equivalence leg was graded; §5 is the exploratory substitute.
“Contamination-proof / newer than every index”	“ Post-Brain-index ”: held out from the Brain’s indexes only; 51/100 golds were exposed in arms C/E’s checkout at run time. The leak’s direction favored C/E, and D’s edge over E is strongest on unexposed tasks (36.7% vs. 8.2%, $p = 0.0026$) — but D-vs-C is a null on that stratum.
“Arm E worst at 16%; unjoined tool volume actively harmful”	E’s 31 no-output rows were session-limit 429s. Repaired, E is mid-pack at 30.0%, indistinguishable from C.
“WF sets the best rows we know of”	A post-hoc, benchmark-informed union+manual condition obtained the highest rows we observed (QR 0.885, MPR 0.557); clustered CIs exclude the single-arm baselines, but the manual was tuned on the same eval queries.

v1 claim (retired)	v2 replacement (supported)
“Adding the join never made the task more expensive per success”	Deleted. Cost figures are descriptive; tokens-to-solve was never computed; SorryDB’s 10-vs-9 is $p = 1.0$.
brain_cell 63% failure “in the Tier-1 traces”	The figure is from the v2 W-arm eval runs (482 calls, 63.3% failed).

What survives cleanly, with its register: the **hallucination collapse** (D 6.8% vs. 17.7–21.2% fresh, post-repair; checked-and-cited-anyway 13% vs. 30–35%) and its promotion to an independent existence-verifier finding [preregistered-modified + exploratory traces]; the **memorization quantification** (arm A 59.8% → 20.0%) [preregistered-modified]; the **system-mode gap** (0.036 single-shot vs. 0.816 agentic on QR) — an API deficiency, not a content deficiency [exploratory]; the **F-matches-SOTA baseline** on MPR (0.453 vs. 0.461) and the **concept≠premise boundary** (W −18pp) [exploratory]; the **snapshot-rot finding** (45% of pinned tasks unreachable in ~6 months, invisible to HEAD-based liveness checks) [exploratory]; and the **tools-vs-no-tools proving gap** on SorryDB [exploratory, 10 clusters].

9 Corrections and response to external review 1

The review (docs/research/review/REVIEW-1.md, received 2026-07-27) raised eight concerns. We verified every factual claim against the repo before acting (verification-of-review-1.json); the table gives concern → verification outcome → action taken in this edition.

#	Concern	Verification	Action in v2
1	Tier-1 “success” does not measure semantic correctness (equivalence grader is a stub)	CONFIRMED (the “misleading” characterization graded PARTIAL — v1’s body disclosed the gap; its abstract did not)	Metric renamed grounded typecheck rate everywhere including the abstract; blind uncalibrated-judge pass launched as the exploratory substitute (§5); human calibration on the roadmap
2	D vs. E does not isolate the join (D alone holds decl_exists, which optimizes a component of the score; 715 calls on fresh)	CONFIRMED	“The join carries the effect” retired; claims rewritten to “the bundled condition”; decl_exists promoted to an independent finding (§4.6); 2 × 2 factorial is roadmap item 1
3	Arm E failed its manipulation check (4 informal calls vs. B’s 345)	CONFIRMED	Disclosed in §3; yoked single-interface control on the roadmap
4	The fresh set was not isolated from formal-search sources (mutable check-out at 61a5e4f338; live Loogee unshot)	CONFIRMED (64/100 basenames in-tree; 51 own-module)	Set renamed post-Brain-index ; exposure measured per-task and outcomes stratified (§4.4); frozen-snapshot protocol on the roadmap
5	WF is test-set tuned (manual written after and from the same eval’s N/F/W results)	CONFIRMED (commit times 22:27 → 00:30 → 01:15; every bracketed manual figure reproduces from the eval aggregates)	WF labeled post-hoc, benchmark-informed at every mention including the abstract; the manual’s same-eval-set provenance stated plainly (§6); the v1 misattribution of the 63% brain_cell figure to Tier-1 corrected; bare-union arm slotted; dev-set manual freeze on the roadmap
6	Statistical independence and execution (810 QR rows are 171 declarations; sequential arm blocks; advisory turn budget; single seed; no CIs)	CONFIRMED	Declaration-clustered QR reanalysis (§6.1); Wilson/bootstrap CIs on every rate; RD + CI on paired effects; sequential blocks, overrun counts (C 50/D 38/E 32, max 88), and single-seed status disclosed in §3; row-level tests survive only as uncorrected footnotes
7	The preregistered primary experiment was not completed, and the deviations list did not say so	CONFIRMED	The complete execution table (§3) — every component, status, inferential consequence; the three-register labeling throughout; the zero-confirmatory statement in the abstract and §3
8	SorryDB provides little evidence about the join (no W-only/bare-union arm; 94% formal routing; missing verdicts; unstable cost-per-proof)	CONFIRMED	All 203 candidate verdicts completed (9682b3b1); WF-vs-F stated as indistinguishable ($p = 1.0$; 34% of repo-bootstrap resamples ≤ 0); repo-clustered CIs; the cost-per-success claim deleted

Two of the review’s factual claims did not survive verification, recorded here with appreciation rather than triumph — the review’s error rate across 31 checked claims was two. First, MathlibMPR does not

cluster by PR: the data file has 69 tasks and 69 distinct PRs, one task each, so the MPR analysis needed no cluster correction (the refuted claim). Second, arm E’s 31 declaration-less fresh runs were not a behavioral response to interface overload — they were contiguous session-limit 429 errors at the tail of E’s sequential block. That correction cuts both ways: it removes the review’s “interface overload” reading, and it removes v1’s “unjoined volume is harmful” claim along with the error-inflated headline $p = 2.4 \times 10^{-5}$ (which appears in this edition only in this paragraph, as the corrected artifact it is; the repaired contrasts are §4.2’s).

10 Roadmap

Adopted from the review’s priority list, in its order:

1. **2×2 factorial** — $\{\text{join, no join}\} \times \{\text{decl_exists, none}\}$ on a truly frozen post-cutoff set, same tool counts and schemas; the experiment that turns §4.6’s attribution question into a measurement.
2. **Yoked unjoined control** — one integrated interface returning equally concise informal and formal panels without declaring correspondences, plus a preregistered manipulation criterion for actual dual-corpus use.
3. **Manual freeze on development data** — split MathlibQR by its 171 target declarations, freeze the manual on a dev split, evaluate on unseen targets; run bare-union, F+manual, W+manual, union+manual.
4. **Human judge calibration** — the preregistered 50-item hand-graded set, judge–human agreement reported, before any faithfulness number leaves the exploratory register.
5. **In-loop compiler proving** — the SorryDB follow-up with identical build access per arm; external validity the one-shot protocol lacks.
6. **Multi-seed, multi-model** — ≥ 3 seeds per task and a second model family; no variance estimate exists anywhere in the current data.
7. **Frozen search snapshots** — read-only pinned checkout predating every test theorem, locally snapshotted Google index, per-run hashes of tree, index, Brain snapshot, prompts, and manifests.

11 Data availability

Everything needed to recompute this report lives in the private preservation repository **Deicyde/wikilean-bridge-experiment** (report at the root as `README.md` and under `report/`). Provenance commits refer to the WikiLean working repository.

Path (preservation repo)	Contents	Key provenance commits (WikiLean)
<code>docs/research/BRIDGE-EXPERIMENT.md</code>	preregistration incl. deviations 1–7	0d36f266 (2026-07-16)
<code>docs/research/BRIDGE-RESULTS.md</code>	Tier-1 + retrieval results log	53f5cb31, daac6107
<code>docs/research/review/</code>	external review 1 verbatim + independent claim-by-claim verification (29 confirmed / 1 partial / 1 refuted)	f97e67e4
<code>bench/*.py,</code> <code>bench/arms/</code>	<code>bench/README.md,</code> Tier-1 harness: runner, scorer, REPL typecheck rig, trace analysis, arm manifests	068abe34, e554902e, 6c2ca704, 31b95caa, 79ac3dfe
<code>bench/data/bridge_tasks.jsonl</code> (+.stats)	371 ProofNet# tasks; source pin + MIT licence	e554902e, ec2a9e11
<code>bench/data/fresh_tasks.jsonl</code> (+.stats)	100 post-Brain-index tasks + determinacy annotations + the (Brain-scoped) held-out check	df5dbf92, a0d45103
<code>bench/data/bridge_summary.json</code> <code>bench/data/runs/</code>	v1 Tier-1 scored summary (pre-repair) all 2,355 Tier-1 run rows, incl. the 31 repaired E rows	53f5cb31 19a90209 (repair driver)
<code>bench/data/runs_E_fresh_429_archive/</code> <code>bench/data/runs_devleak_2026-07-18/</code>	the 31 original 429-errored E rows, byte-preserved the 120 quarantined skill-leak dev runs (deviation 6)	f97e67e4 f89e7a41

Path (preservation repo)	Contents	Key provenance commits (WikiLean)
bench/analysis/	v2 corrective statistics: tier1_reanalysis.{py,json,md}, fresh_exposure.{py,json,md}, retrieval_clustered.{py,json,md}, rerun_E_fresh.py, score_e31_v2.py → bridge_summary_v2.json + part1_fresh100_v2.{json,md}, judge_fresh_run.py / judge_fresh_summary.py	64c48052, 19a90209
bench/v2/	v2 harness + AGENT_MANUAL.md	c7629584, 21032c06
bench/v2/data/	MathlibQR/MPR (LeanSearch-v2 @94f4888cbaf9, CC BY 4.0), SorryDB-2601 split, tasks_frozen.jsonl	c7629584, 87638cfc, db679e8d
bench/v2/runs/	every v2 run row with full gzipped stream-json transcripts (QR 810×4, MPR 69×4, SorryDB 508 rows) + verify.jsonl — all 203 frozen- candidate kernel verdicts, complete	3664aa3d, daac6107, 382e51bf, 1fe21cca, 9682b3b1

Recomputation entry points: bench/score_bridge.py (Tier-1 grading), bench/analysis/tier1_reanalysis.py + score_e31_v2.py (§4), bench/analysis/fresh_exposure.py (§4.4), bench/analysis/retrieval_clustered.py (§6.1–6.2, §7 uncertainty), bench/trace_analysis.py (§4.5), bench/v2/score_retrieval.py (§6 point estimates), bench/v2/verify_sorrydb.py (§7 verdicts), bench/analysis/judge_fresh_summary.py (§5, pending).

External sources cited (fetched and verified during the v2 design sweep): TheoremGraph arXiv:2606.25363 · LeanSearch arXiv:2403.13310 · LeanSearch-v2 arXiv:2605.13137 + github.com/frenzymath/LeanSearch-v2 · LeanExplore arXiv:2506.11085 · SorryDB arXiv:2603.02668 + sorrydb.org · FATE arXiv:2511.02872 · miniCTX arXiv:2408.03350 · LeanDojo arXiv:2306.15626 · LeanAgent arXiv:2410.06209 · Numina-Lean-Agent arXiv:2601.14027 · miniF2F-v2 arXiv:2511.03108 · ProofNet#/ProofNetVerif arXiv:2406.07222 · benchmark-faults survey arXiv:2606.29493.

A The agent manual

*The following is bench/v2/AGENT_MANUAL.md, the briefing prepended verbatim to every WF-arm prompt. It is reproduced unchanged as a study artifact — and, per §6, it is a **post-hoc, benchmark-informed** artifact: its bracketed figures were measured on the same MathlibQR/MPR evaluation queries the WF arm was then scored on.*

Tool manual: searching Mathlib with the Brain + formal search

You have two complementary toolkits. Every claim in this manual is measured from ~5,500 logged tool calls across 3,500 benchmark runs (Tier-1 + Bridge v2); numbers in brackets are those measurements.

A.1 The mental model

- **The Brain (wikibrain, 8 tools)** is a *curated knowledge graph* joining informal mathematics (Wikipedia/Wikidata concepts, external databases) to Mathlib declarations. It knows what things *mean* and how concepts relate — but it only “atomizes” declarations that some concept, annotation, or tag claims (~12k of Mathlib’s declarations). It is a map, not the territory.
- **Formal search (3 tools)** operates on *all* of Mathlib directly: type-pattern search (loogle), source grep, source reading. It knows everything that exists but nothing about what it means informally.

Rule of thumb: **route by what you’re holding**. Holding a concept described in words → start with the Brain. Holding a type shape or name fragment → start with formal search. Always finish with `decl_exists` on every name you intend to output.

A.2 Formal search tools

A.2.1 loogle

Type-pattern and constant search (the public `loogle.lean-lang.org` service).

- USE for: “a lemma whose statement looks like `_ * _ ≤ _ * _`”, searches by involved constants (`Real.sqrt`, `tsum`), name substrings.
- Strengths: covers ALL of Mathlib; $\sim 0\%$ call errors [578 calls, 0 errors]; the best premise-finding tool — the loogle+grep toolkit matched a specialist premise retriever [group-recall@10 0.453 vs 0.461 SOTA].
- Weaknesses: needs Lean syntax fluency; a wrong pattern silently returns unrelated hits; small result payloads mean you often need `decl_read` next.

A.2.2 decl_grep

Regex search over the Mathlib source checkout (ripgrep).

- USE for: name fragments (“cantor”), notation, docstring phrases, finding the file a declaration lives in.
- Strengths: the workhorse [1,206 calls, 0% errors]; catches things loogle’s type index can’t (comments, docstrings, naming conventions).
- Weaknesses: lexical only — if the concept’s name doesn’t appear in source, grep cannot find it (descriptive queries like “special case of X” score worst for grep-based agents [nDCG 0.384 vs the Brain’s 0.523]).

A.2.3 decl_read

Read declaration source (with surrounding context) from the checkout.

- USE for: verifying a candidate actually says what you think BEFORE citing it; getting exact signatures and hypotheses.
- Strengths: ground truth for any decl in Mathlib [397 calls, 0% errors].
- Weaknesses: you need the file/name first — it is a confirmation tool, not a discovery tool.

A.3 Brain tools

A.3.1 brain_bridge — THE ENTRY POINT for informal→formal

Free text in, existence-verified declarations + owning atoms out, with signatures, import lines, and one-hop dependency context.

- USE for: “the concept described as <words>” — especially nicknames and special-case descriptions, where the Brain’s curated bonds beat grep [special_case nDCG 0.523 vs 0.384; nickname 0.779 vs 0.757].
- CRITICAL LIMITATION (measured): its free-text matching is label/alias anchored, NOT semantic. A single call with a descriptive paraphrase usually misses [single-shot benchmark: 3.6% R@10 vs 82% for an agent that reformulates]. **Never accept one miss as the answer: reformulate 2–4 times** — try the concept’s canonical name, a synonym, the head noun alone. 8.9% of calls return `match:"none"` [727 calls]; that response includes nearest-candidate suggestions — read them.
- The `hits` list is existence-verified; bond quality matters: `exact` means the decl IS the concept; `formalizes`/weaker bonds are nearby.

A.3.2 brain_search

Fuzzy label + alias search returning atoms (cells).

- USE for: resolving a name you half-know into an atom id; browsing what the Brain calls something [400 calls, 0% errors — the reliable fallback when `brain_bridge` misses].
- Weaknesses: returns atoms, not ranked decls — follow with `brain_cell` on the winning atom.

A.3.3 brain_cell

The full atom card: Lean code, Wikidata description, DB snippets, organs.

- USE for: everything known about ONE object after bridge/search resolved it.
- **THE #1 MISUSE [63% of 482 calls failed]: calling it with a bare decl:Mathlib:Name for an arbitrary declaration.** The Brain only has atoms for concept-joined decls; “unresolvable key” means *no atom owns this decl*, NOT that the decl doesn’t exist. For arbitrary decls use decl_exists (works for all of Mathlib) and decl_read for source. Call brain_cell only with ids that came OUT of a Brain tool.

A.3.4 decl_exists — THE DISCIPLINE TOOL

Batch existence check for declaration names (backed by the full oracle — covers ALL of Mathlib, unlike brain_cell).

- **USE ALWAYS, on every name you are about to output.** This is the measured difference between grounded and hallucinated citations: agents that verified before citing cut hallucinated-but-cited-anyway to 13% vs 33% for agents relying on fuzzy search results alone. It is cheap [1,473 calls, 0.1% errors] and batched — check 10 names in one call.
- Also returns verified-rename suggestions for stale names.

A.3.5 brain_neighborhood

The synapse graph around an atom (depends/links/relates, cursored).

- USE for: “what connects to X” — finding the connecting lemma between two concepts, walking dependencies.
- Weaknesses: needs a valid atom id [24.8% of calls errored — same unresolvable-key trap as brain_cell; resolve the atom first].

A.3.6 brain_transfer / brain_snippets / brain_filter

Narrower tools: label↔decl jumps for KNOWN labels (transfer), stored source snippets (snippets), facet enumeration (filter). High misuse rates when fed non-atom ids [40%/57% errors]. Prefer brain_bridge/brain_search entry; reach for these only on ids the Brain already returned.

A.4 Routing playbook

You are holding	Do this
A concept in words	brain_bridge → (miss?) reformulate → brain_search → brain_cell
A special case / nickname	brain_bridge (the Brain’s best terrain)
A type shape	loogle
A name fragment	decl_grep
Premises for a proof	loogle + decl_grep FIRST [0.453 vs 0.272], Brain for concept grounding
A candidate name to cite	decl_exists (batch, always) → decl_read if the statement matters
An atom id from any Brain tool	brain_cell / brain_neighborhood freely
A bare decl name needing info	decl_exists + decl_read — NOT brain_cell

A.5 Anti-patterns (each one measured in prior runs)

1. **Citing unverified names.** 33% of hallucinated citations came from agents that had search evidence in front of them and cited a wrong name anyway. decl_exists is binary and kills this.
2. **One query, then giving up.** The single-call miss rate on descriptive queries is ~96%; agents that reformulate reach ~82%. Iteration IS the algorithm.
3. **brain_cell as a generic decl inspector.** 63% failure rate. Brain atoms ≠ all of Mathlib.
4. **Answering outside the requested format.** 143/810 no-tool runs were scored 0 for exactly this. Re-read the output contract before replying.

5. **Inventing tool names** (e.g. `logue`, `lookie`). Use the names above.

DRAFT